Practice Set 1 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Classify testicular tumors and describe management of seminoma (LAQ)
2. Q2. Define algorithm for evaluation of antenatally diagnosed unilateral HN in male fetus. What are the current rate of prenatal fetal anomalies (LAQ)
3. Q3a. ANH (SAQ)
4. Q3b. Role of UDS in SUI (SAQ)
5. Q3c. Retrocecal ureter (SAQ)
6. Q3d. Spermatogenesis (SAQ)
7. Q3e. Enuresis (SAQ)
8. Q3f. Female sexual dysfunction (SAQ)

Practice paper compiled from the question bank.

Q1. Classify testicular tumors and describe management of seminoma

Type: LAQ (25 marks) (stage I seminoma also answered in 2018-Summer Paper 2, Q2)

Introduction

- Over 95% of testicular tumours are GCTs (germ cell tumours). Seminoma is 52–56% of GCNIS (germ cell neoplasia in situ)-derived GCTs, peaks in the fourth–fifth decade.
- **Risk factors:** cryptorchidism, previous contralateral GCT (twelvefold), family history, GCNIS, White race, infertility.
- All postpubertal GCTs except spermatocytic tumour carry **isochromosome 12p**.

WHO (World Health Organization) classification (2016, retained 2022)

Group	Tumours
1. GCNIS-derived GCTs	GCNIS; **seminoma**; **NSGCT (non-seminomatous GCT)**: embryonal carcinoma, postpubertal yolk sac tumour, choriocarcinoma and other trophoblastic tumours, postpubertal teratoma, teratoma with somatic-type malignancy; mixed GCT
2. Non-GCNIS-derived GCTs	**Spermatocytic tumour** (formerly spermatocytic seminoma); prepubertal teratoma, dermoid and epidermoid cyst; prepubertal yolk sac tumour
3. Sex cord–stromal tumours (<5%)	Leydig cell, Sertoli cell (including large cell calcifying), granulosa cell (adult, juvenile), fibroma–thecoma, mixed
4. Mixed germ cell–sex cord stromal	Gonadoblastoma (DSD [difference of sex development])
5. Others	Lymphoma (commonest testicular tumour in older men), rete testis and collecting duct tumours, adnexal and paratesticular tumours, metastases

- Any NSGCT element, however small, makes the tumour an NSGCT and is treated as such.
- **Seminoma pathology:** sheets of clear cells with lymphocyte-rich fibrovascular septa. Immunohistochemistry: CD117 and PLAP (placental alkaline phosphatase) positive, CD30 negative. Syncytiotrophoblasts (hCG [human chorionic gonadotropin]) in about 15%. "Anaplastic seminoma" is no longer recognised.
- **Spermatocytic tumour** (<1%): older men, no GCNIS, no i(12p), benign unless sarcomatous.

Evaluation and staging

- Scrotal ultrasound; AFP (alpha-fetoprotein), β-hCG and LDH (lactate dehydrogenase) before and after orchidectomy. **A raised AFP means NSGCT**, whatever the histology.
- **Radical inguinal orchidectomy** with early cord control at the internal ring; never trans-scrotal. Offer sperm banking and prosthesis.
- CT (computed tomography) of chest, abdomen and pelvis; TNMS (tumour, node, metastasis, serum markers) staging.
- About 80% of seminomas are clinical stage (CS) I; 15–20% CS II (nodes IIA ≤2 cm, IIB 2–5 cm, IIC >5 cm).

Management of seminoma by stage

CS I (cure by orchidectomy alone 80–85%; cancer-specific survival about 100% with any option)

	Surveillance (preferred)	Carboplatin	Radiotherapy
Method	Examination, markers, CT for at least 5 years	1 cycle **AUC (area under the curve) 7**, dose = 7 × (GFR [glomerular filtration rate] + 25) mg	**20 Gy** para-aortic strip
Relapse	14–20%	About 5% (1 cycle)	3–5%
Drawback	Late relapses (10–20% after 4 years)	Underdosing; late effects unknown	Second cancers, GI (gastrointestinal) toxicity

- **Relapse risk factors:** tumour **>4 cm** and **rete testis invasion** (relapse 12%, 16%, 32% with 0, 1, 2 factors). LVI (lymphovascular invasion) is not a factor in seminoma.
- **Trials:** MRC TE19/EORTC (Medical Research Council/European Organisation for Research and Treatment of Cancer) 30982: one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE18: 20 Gy equals 30 Gy.
- Surveillance relapses are 84–100% retroperitoneal and almost all salvaged.

CS IIA–B

- **Non-bulky (≤3 cm):** dog-leg radiotherapy with nodal boost (25–30 Gy in the chapter; EAU [European Association of Urology] uses 30 Gy for IIA and 36 Gy for IIB), or chemotherapy, or **primary RPLND (retroperitoneal lymph node dissection)** (SEMS trial).
- **Bulky (>3 cm) or multiple nodes:** BEP (bleomycin, etoposide, cisplatin) ×3 or EP (etoposide, cisplatin) ×4.

CS IIC–III: chemotherapy by IGCCCG (International Germ Cell Cancer Collaborative Group)

Risk	Seminoma definition	5-year survival	Regimen
Good (90%)	Any primary, **no NPVM (non-pulmonary visceral metastases)**, normal AFP, any hCG/LDH	86%	BEP ×3 or EP ×4
Intermediate (10%)	NPVM present	72%	BEP ×4
Poor	**None for seminoma**	–	–

- Carboplatin alone is inferior in advanced disease.

Post-chemotherapy residual mass

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Residual mass after chemotherapy
 ├─ <3 cm ──────────────► observe (necrosis about 90%)
 └─ >3 cm ─► FDG-PET (fluorodeoxyglucose positron emission tomography) at ≥6 weeks
              ├─ negative ► observe
              └─ positive ► resection (full template PC-RPLND) or salvage
```

- Surgery is difficult because of desmoplasia; complete resection only 58–74%.

Relapse and follow-up

- After surveillance or radiotherapy: radiotherapy for small retroperitoneal relapse, otherwise first-line BEP. After chemotherapy: VeIP (vinblastine, ifosfamide, cisplatin) or high-dose chemotherapy.

- Surveillance schedule: review every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly. CT detects >85% of relapses.

Recent advances

- Serum **miR-371a-3p** microRNA: sensitive marker of residual disease and relapse.
- MRI (magnetic resonance imaging) for surveillance to cut radiation; primary RPLND for low-volume stage II.

Diagram to draw: WHO classification tree (GCNIS-derived vs non-GCNIS-derived vs sex cord–stromal) and the residual-mass flowchart.

> **Key points to remember:**
>
> - WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
> - CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
> - Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
> - No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

Q2. Define algorithm for evaluation of antenatally diagnosed unilateral HN in male fetus. What are the current rate of prenatal fetal anomalies

Type: LAQ (25 marks) (antenatal LUTO management also answered in 2021-Summer Paper 3, Q3b)

Assumed meaning: HN = hydronephrosis, now called UTD (urinary tract dilation).

Introduction and rates of prenatal anomalies

- Major congenital anomalies occur in about **2–3% of births**. Urinary tract anomalies are among the commonest detected prenatally (commonly quoted as 20–30% of all prenatal anomalies).
- **UTD is the second commonest prenatal anomaly after cardiac defects, in 1–3% of pregnancies.**
- Most is mild: mild 57–88%, moderate 10–30%, severe 1.5–13% of UTD.
- Screening US (ultrasound) at 18–20 weeks detects dilation well (sensitivity about 90%), but predicts diagnosis poorly. Fetal urine production begins at about 14 weeks; amniotic fluid is mainly fetal urine after 16 weeks.

Causes (SFU [Society for Fetal Urology] consensus)

Cause	Incidence
Transient/physiological dilation	50–70%
UPJ (ureteropelvic junction) obstruction pattern	10–30%
VUR (vesicoureteral reflux)	10–40%
UVJ (ureterovesical junction) obstruction/megaureter	5–15%
MCDK (multicystic dysplastic kidney)	2–5%
Duplex system, ureterocele, ectopic ureter	5–7%
LUTO (lower urinary tract obstruction: PUV [posterior urethral valves], prune-belly)	0.1–0.5%

- Chance of postnatal pathology rises with grade: about 12% (mild) to 88% (severe). UPJ obstruction and PUV track grade; **VUR does not**.

Grading systems

APD (anteroposterior renal pelvic diameter): transverse image at the hilum.

Grade	Second trimester	Third trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm
Severe	>10 mm	>15 mm

SFU (postnatal only): 1 pelvis barely split; 2 pelvis and major calyces; 3 minor calyces uniformly dilated; 4 plus parenchymal thinning.

UTD consensus 2014: six parameters: APD, central/peripheral calyces, parenchymal thickness, parenchymal appearance, ureter, bladder (plus amniotic fluid prenatally).

Prenatal	A1 low risk	A2–3 increased risk (any one)
APD 16–27 weeks	4 to <7 mm	≥7 mm

Prenatal	A1 low risk	A2–3 increased risk (any one)
APD ≥28 weeks	7 to <10 mm	≥10 mm
Others	Central calyces only; all normal	Peripheral calyces, abnormal parenchyma, ureter or bladder, unexplained oligohydramnios

Prenatal algorithm (unilateral UTD, male fetus)

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Anomaly scan: unilateral UTD
 → Detailed scan: contralateral kidney, ureter, bladder wall and emptying,
   posterior urethra (keyhole), amniotic fluid, other anomalies
 ├─ Bilateral / thick bladder / keyhole / oligohydramnios ─► LUTO pathway
 │    (fetal medicine; karyotype; Ruano staging; possible shunt)
 ├─ A1 ─► repeat US after 32 weeks
 └─ A2–3 ─► US every 4 weeks to delivery
 → Normal delivery at local centre; no fetal intervention for unilateral UTD
 → Postnatal US after 48 hours (earlier if solitary kidney or suspected LUTO)
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- Repeat scans because the lesion may progress and a second anomaly is more likely. **Fetal MRI** only for complex anatomy.
- Counsel the parents: most cases resolve; prognosis with a normal opposite kidney is excellent.

Postnatal algorithm

- **First US after 48 hours**, because neonatal dehydration under-reads dilation. Postnatal normal: APD <10 mm, no calyceal or ureteral dilation.

	P1 (SFU 1–2)	P2 (SFU 3)	P3 (SFU 4)
Findings	APD 10–15 mm, central calyces	APD ≥15 mm, peripheral calyces or dilated ureter	Abnormal parenchyma or bladder
Repeat US	3–6 months	Individualised	1 month
Antibiotic prophylaxis	No	Discretion; **yes if ureter >7 mm**	Yes
VCUG (voiding cystourethrogram)	No	Discretion	Yes
MAG3 (mercaptoacetyltriglycine) renogram	No	Discretion	If persistent

- **Male-specific points:** VCUG confirms a normal posterior urethra. **Circumcision** protects (NNT [number needed to treat] 10). UTI (urinary tract infection) risk: female sex, uncircumcised boys, high grade, dilated ureter.
- Prophylaxis: amoxicillin or a first-generation cephalosporin from birth; nitrofurantoin and trimethoprim only after 8 weeks.
- MAG3: DRF (differential renal function) <40%, T½ >20 minutes, or delayed retention suggest obstruction. MRU (magnetic resonance urography) for anatomy; DMSA (dimercaptosuccinic acid) for MCDK or duplex function.
- **Surgery indications (UPJ):** rising APD, DRF <40% with poor drainage, falling DRF, symptoms (UTI, stone, pain). Megaureter surgery: DRF <40%, febrile UTI; defer reimplant beyond 1 year.
- Outcome: SFU 1 resolves in nearly all by 2.5 years; SFU 2 about 95% by 4 years.

Bilateral dilation and PUV: how they differ

- Bilateral hydroureteronephrosis, thick bladder, keyhole sign, oligohydramnios = LUTO until proved otherwise.
- At birth: 5 Fr feeding tube, prophylaxis, creatinine, early VCUG; prenatal intervention improves survival but not renal function (PLUTO).

Diagram to draw: UTD A1/A2–3 prenatal and P1–P3 postnatal flowchart.

Key points to remember:

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Transient dilation 50–70%; UPJ obstruction commonest surgical cause.
- First postnatal US after 48 hours; A2–3 scan every 4 weeks.
- P1 no prophylaxis or VCUG; P3 prophylaxis, VCUG, MAG3.

Q3a. ANH

Type: SAQ (10 marks)

See Practice Set 1 Paper 3, Q2 (full answer).

Key points to remember (SAQ angle):

- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- Commonest cause transient dilation; VUR is not predicted by grade.
- Risk-based care: no prophylaxis or VCUG for P1; prophylaxis if ureter >7 mm or P3.

Q3b. Role of UDS in SUI

Type: SAQ (10 marks) (SUI evaluation also answered in 2019-Summer Paper 3, Q2)

Introduction

- UDS (urodynamic studies) assess storage and voiding. In SUI (stress urinary incontinence) the key question is: **will the result change the operation?** Pure, demonstrable SUI gains little.

What UDS shows in SUI

Parameter	Method and meaning
USI (urodynamic stress incontinence)	Leak with raised abdominal pressure, no detrusor contraction
ALPP (abdominal leak point pressure)	Standing, from 150 mL, Valsalva or cough; **<60 cmH2O = ISD (intrinsic sphincter deficiency)**, 60–90 equivocal, ≥90 little ISD (McGuire 1993)
MUCP (maximum urethral closure pressure)	Urethral pressure profile; ≤20 cmH2O suggests ISD; does not diagnose or grade SUI
Filling cystometry	DO (detrusor overactivity), compliance, capacity; cough-induced DO mimics SUI
Pressure–flow study	Voiding dysfunction, poor contractility (retention risk after sling)
Prolapse reduction	Occult SUI

- If no leak with the catheter, repeat with it removed.

Evidence

- **ValUE trial (2012), 630 women** with uncomplicated SUI: office evaluation alone was **non-inferior** to office evaluation plus UDS (success 77.2% vs 76.9%).
- **Dutch VUE study:** immediate midurethral sling was non-inferior to UDS-tailored treatment.
- **SISTEr:** UDS did not predict outcome.

Guidelines

- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023:** UDS may be omitted in the index patient with demonstrable SUI.
- **UDS advised in non-index patients:** previous anti-incontinence or prolapse surgery; urgency-predominant mixed incontinence; raised PVR (post-void residual) or voiding dysfunction; SUI not demonstrated; mismatch between symptoms and findings; neurogenic bladder.
- **EAU (European Association of Urology):** no routine UDS for uncomplicated SUI; do not use leak point or urethral pressure to grade severity.

Men

- Before AUS (artificial urinary sphincter) or sling for post-prostatectomy incontinence: confirms USI, finds DO, poor compliance or weak contractility. MASTER trial recruited men with USI.

Diagram to draw: cystometrogram trace with abdominal pressure spike and leak (ALPP).

Key points to remember:

- ValUE: UDS not needed in uncomplicated SUI.
- ALPP <60 cmH2O = ISD; MUCP ≤20 cmH2O.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.

Q3c. Retrocecal ureter

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 3, Q3d)

Assumed meaning: retrocaval (circumcaval) ureter; "retrocecal" is taken as a typo.

Definition

- The **right ureter** passes **behind the IVC (inferior vena cava)**, then crosses in front of it from medial to lateral.
- Better called **"preureteral vena cava"**: the anomaly is venous; the ureter is trapped.

Embryology

- Fetal veins form a collar around each ureter: supracardinal veins dorsally, subcardinal veins ventrally.
- Normally the right **supracardinal** vein (dorsal to the ureter) forms the infrarenal IVC, so the ureter lies in front.
- If the lumbar part of the ventral **subcardinal** vein persists as the IVC, the ureter is trapped behind it.
- Variant: double right IVC (ureter between limbs).

Incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**.
- Presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis. Associations: horseshoe kidney, left renal anomalies.

Classification (Bateson and Atkinson)

	Type I (low loop), commoner	Type II (high loop)
Crossing	Lower (L3–L4)	Higher, near the renal pelvis
Shape	**"Fishhook"/reverse-J**	Smooth horizontal sickle curve
Obstruction	Moderate–severe, at the iliopsoas edge	Mild or none

Investigations

- **CT urography (computed tomography urography): investigation of choice**; MRI (magnetic resonance imaging) alternative.
- IVU (intravenous urography): fishhook; ureter not seen beyond the J hook. Retrograde pyelography: **S-curve** at L3–L4.
- **Diuretic renogram:** obstruction and split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**; pyeloplasty fails.

Management

- **Non-obstructed (often type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** or ureteropyelostomy. Divide the ureter, transpose it in front of the IVC, spatulated anastomosis over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**.
- **Laparoscopic or robotic** repair now preferred; nephrectomy if non-functioning.
- **Also know:** retroiliac ureter; mimics: gonadal or lumbar vein, iliopsoas tendon.

Diagram to draw: right ureter hooking behind the IVC with fishhook pelvis.

Key points to remember:

- Venous anomaly (persistent subcardinal vein); right side; 1 in 1500; male 3:1.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; anterior transposition with ureteroureterostomy.

Q3d. Spermatogenesis

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 1, Q2)

Definition

- Production of haploid spermatozoa from diploid spermatogonia in the seminiferous tubules: about **64 days** (42–76); about 123 million sperm a day.

Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- Pulsatile **GnRH (gonadotropin-releasing hormone)** → **LH (luteinising hormone)** → Leydig cells → testosterone. **FSH (follicle-stimulating hormone)** → Sertoli cells.
- Feedback: testosterone on LH; **inhibin B** (Sertoli) on FSH.
- **Intratesticular testosterone** is about 50 times serum, held by ABP (androgen-binding protein). Exogenous testosterone suppresses LH and so stops spermatogenesis.

Sertoli cells and blood–testis barrier

- Nurse cells: nourish germ cells; secrete ABP, inhibin, fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** form a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). This protects immunologically "foreign" post-meiotic cells.

Phases

Phase	Events
1. Proliferation	**Ad (dark type A)** stem cells self-renew (GDNF [glial cell line–derived neurotrophic factor] niche); **Ap (pale type A)** divide every 16 days into type B → primary spermatocytes. Daughters remain joined by cytoplasmic bridges
2. Meiosis	Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; synapsis and crossing-over) → secondary spermatocytes → meiosis II → 4 haploid round spermatids
3. Spermiogenesis	No division: acrosome from Golgi, flagellum from centriole, nucleus compacts to 10%, mitochondria form the midpiece, cytoplasm shed
4. Spermiation	Sertoli cell releases sperm into the lumen

- **Cycle, stage, wave:** six stages in humans; cycles arranged as a helical wave, so production is continuous.

Post-testicular maturation

- Epididymal transit 2–12 days: sperm gain **progressive motility** and fertilising ability; stored in the cauda. **Capacitation** and acrosome reaction occur in the female tract.

Clinical relevance

- A semen sample reflects events about 3 months earlier; reassess 3 months after treatment.
- Disruptors: heat (testis normally 2–4 °C below core; varicocele, cryptorchidism), exogenous testosterone, chemotherapy, radiation.
- Defective meiotic recombination causes maturation arrest.

Diagram to draw: seminiferous tubule cross-section with basal and adluminal compartments.

Key points to remember:

- About 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; FSH → Sertoli, LH → Leydig.
- Tight junctions between Sertoli cells form the blood–testis barrier.

Q3e. Enuresis

Type: SAQ (10 marks)

Definition (ICCS [International Children's Continence Society])

- Intermittent urinary incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

Epidemiology

- About **5–10% at 7 years**; spontaneous resolution about 15% a year; about 2% of adults; familial.

Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (reduced night-time ADH [antidiuretic hormone] rise).
2. **Reduced nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep.

- Comorbidities: constipation, sleep-disordered breathing (atrial natriuretic peptide raises urine output), ADHD (attention deficit hyperactivity disorder).

Evaluation

- History: day symptoms, bowels, snoring, fluids, family history.
- **Bladder diary**: daytime voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL in the chapter (ICCS uses [age + 1] × 30); night urine output by weighing nappies.
- Examination: spine, genitalia, abdomen for faecal loading; urinalysis.
- Uroflowmetry, PVR (post-void residual) and ultrasound only for NMNE or before adding anticholinergics. No invasive tests in MNE.

Management

- **Treat daytime LUTS and constipation first** in NMNE.
- Education removes blame; standard urotherapy. Waking, fluid restriction alone and punishment do not work.

	Alarm	Desmopressin
Target	Arousal (conditioning)	Night urine output
Use	Nightly for months; parent wakes child; continue to 14 dry nights	0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed
Response	50–70%; durable	Rule of thirds; immediate but relapses
Caution	Needs motivated family	Restrict evening fluids (hyponatraemia)

- Overlearning (extra evening fluid once dry) reduces alarm relapse.
- **Failure:** switch or combine. **Second line:** add an anticholinergic (oxybutynin) after checking PVR and bowels. **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists; treat sleep apnoea (adenotonsillectomy).

Diagram to draw: triad diagram: nocturnal polyuria, small bladder capacity, arousal failure.

Key points to remember:

- ≥5 years; secondary = after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

Q3f. Female sexual dysfunction

Type: SAQ (10 marks)

Definition and prevalence

- FSD (female sexual dysfunction): persistent desire, arousal, orgasm or pain problems causing **personal distress**.
- About **40%** of women in their lifetime; **HSDD (hypoactive sexual desire disorder) commonest**. Almost half of women with UI (urinary incontinence) have FSD.

Physiology and response models

- Arousal: pudendal afferents, pelvic parasympathetic efferents → **NO (nitric oxide)–cGMP (cyclic guanosine monophosphate)** → engorgement and lubrication. Oestrogen maintains tissues; testosterone drives desire.
- **Masters and Johnson** (linear): excitement, plateau, orgasm, resolution; Kaplan added desire. **Basson** (circular): desire is often responsive.

Classification

DSM-5 (Diagnostic and Statistical Manual, 2013)	ICSM (International Consultation on Sexual Medicine)/ISSWSH (International Society for the Study of Women's Sexual Health)
Female sexual interest/arousal disorder (≥3 of 6 criteria)	HSDD and arousal disorder kept separate (cognitive vs genital arousal)
Female orgasmic disorder	Orgasm disorders
Genito-pelvic pain/penetration disorder	Genital-pelvic pain dysfunction
–	PGAD (persistent genital arousal disorder)

- ≥6 months with distress; lifelong or acquired; generalised or situational.

Causes

- Hormonal (menopause, GSM [genitourinary syndrome of menopause]), drugs (SSRIs [selective serotonin reuptake inhibitors], antihypertensives), diabetes, depression, relationship factors.
- Urological: incontinence, prolapse, sling dyspareunia, cystectomy nerve or clitoral injury.

Evaluation

- Clinician must ask. **PLISSIT** (permission, limited information, specific suggestions, intensive therapy); BSSC-W (Brief Sexual Symptom Checklist for Women) screen; **FSFI (Female Sexual Function Index, 19 items, six domains)**.
- Examination: atrophy, lichen sclerosus, cotton-swab test, pelvic floor tone. Laboratory tests rarely help.

Management

- Treat causes; education; sex therapy (sensate focus, mindfulness).
- **HSDD (premenopausal, FDA [US Food and Drug Administration] approved): flibanserin** 100 mg at bedtime (5-HT1A agonist/5-HT2A antagonist; alcohol caution); **bremelanotide** 1.75 mg subcutaneously on demand (melanocortin agonist). Postmenopausal: transdermal testosterone (off-label).

- **Arousal:** testosterone; PDE5 (phosphodiesterase type 5) inhibitors only with neurovascular disease.
- **Orgasm:** education, directed masturbation.
- **Pain/vestibulodynia:** pelvic floor physiotherapy, dilators, biofeedback, topical lidocaine, vaginal oestrogen; vestibulectomy if refractory.
- **Also know:** ospemifene, intravaginal prasterone (menopausal dyspareunia).

Diagram to draw: Basson circular sexual response model.

> **Key points to remember:**
>
> - About 40% prevalence; HSDD commonest; distress required.
> - DSM-5 merges desire and arousal, dyspareunia and vaginismus.
> - Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

Rapid Revision Sheet – Practice Set 1 Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Classify testicular tumors and describe management of seminoma (LAQ)

- WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
- CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
- Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
- No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

Q2. Define algorithm for evaluation of antenatally diagnosed unilateral HN in male fetus. What are the current rate of prenatal fetal anomalies (LAQ)

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Transient dilation 50–70%; UPJ obstruction commonest surgical cause.
- First postnatal US after 48 hours; A2–3 scan every 4 weeks.
- P1 no prophylaxis or VCUG; P3 prophylaxis, VCUG, MAG3.

Q3a. ANH (SAQ)

- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- Commonest cause transient dilation; VUR is not predicted by grade.
- Risk-based care: no prophylaxis or VCUG for P1; prophylaxis if ureter >7 mm or P3.

Q3b. Role of UDS in SUI (SAQ)

- ValUE: UDS not needed in uncomplicated SUI.
- ALPP <60 cmH2O = ISD; MUCP ≤20 cmH2O.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.

Q3c. Retrocecal ureter (SAQ)

- Venous anomaly (persistent subcardinal vein); right side; 1 in 1500; male 3:1.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; anterior transposition with ureteroureterostomy.

Q3d. Spermatogenesis (SAQ)

- About 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; FSH → Sertoli, LH → Leydig.
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Q3e. Enuresis (SAQ)

- ≥5 years; secondary = after 6 months dry.
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Q3f. Female sexual dysfunction (SAQ)

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